Project Documentation 2

(Data Processing, Feature Engineering and Analysis)

SaaS Product Analytics Project Documentation

📄 DOCUMENT 1: DATA PROCESSING, FEATURE ENGINEERING & ANALYSIS

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1. Project Title

SaaS Product Analytics & Churn Analysis System

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2. Objective

The objective of this project is to analyze SaaS customer behavior and answer key business questions such as:

- Which customers are likely to churn?
- Which features drive engagement?
- How does support impact retention?
- Which segments generate the most revenue?

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3. Dataset Overview

Worked with multiple datasets:

- Accounts
- Subscriptions
- Feature Usage (event-level)
- Support Tickets
- Churn Events

These datasets contained both customer-level and event-level information.

```
[63] accounts.columns

Index(['account_id', 'account_name', 'industry', 'country', 'signup_date',
       'referral_source', 'plan_tier', 'seats', 'is_trial', 'churn_flag'],
      dtype='object')
```

```
[64]
subscriptions.columns

Index(['subscription_id', 'account_id', 'start_date', 'end_date', 'plan_tier',
       'seats', 'mrr_amount', 'arr_amount', 'is_trial', 'upgrade_flag',
       'downgrade_flag', 'churn_flag', 'billing_frequency', 'auto_renew_flag'],
      dtype='object')
```

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4. Data Cleaning

Steps Performed:

- Converted date columns into proper datetime format
- Handled missing values
- Standardized column names and formats

Example Code:

subscriptions["start_date"] = pd.to_datetime(subscriptions["start_date"])

```
[41]
subscriptions.head()
subscriptions.info()
subscriptions.describe()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 14 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   subscription_id    5000 non-null   object
 1   account_id         5000 non-null   object
 2   start_date         5000 non-null   datetime64[ns]
 3   end_date           486 non-null    datetime64[ns]
 4   plan_tier          5000 non-null   object
 5   seats              5000 non-null   int64
 6   mrr_amount         5000 non-null   int64
 7   arr_amount         5000 non-null   int64
 8   is_trial           5000 non-null   bool
 9   upgrade_flag       5000 non-null   bool
 10  downgrade_flag     5000 non-null   bool
 11  churn_flag         5000 non-null   bool
 12  billing_frequency  5000 non-null   object
 13  auto_renew_flag    5000 non-null   bool
dtypes: bool(5), datetime64[ns](2), int64(3), object(4)
memory usage: 376.1+ KB
```

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5. Data Merging

All datasets were merged using:

- account_id

- subscription_id

Output:

- Created master dataset
- Shape: ~147,000 rows × 46 columns

```
[45]: master_df.shape

[45]: (147896, 46)
```

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6. Feature Engineering (Core Step)

Event-level data was transformed into customer-level aggregated data.

Features Created:

- total_feature_usage
- avg_usage_duration
- total_errors
- total_tickets
- avg_satisfaction
- total_mrr, total_arr

Output:

- Final dataset: (500 rows × 20 columns)

```
analytics_df.shape
analytics_df.head()
```

	account_id	industry	country	seats	is_trial	churn_flag	total_feature_usage	avg_usage_duration	total_errors	beta_feature_usage	total_tickets	avg_resolution_tir
0	A-2e4581	EdTech	US	9	False	False	2140.0	2769.800000	152.0	16	220	23.0000
1	A-43a9e3	FinTech	IN	18	False	True	1065.0	2889.600000	42.0	6	105	38.0000
2	A-0a282f	DevTools	US	1	False	False	4926.0	3026.626506	288.0	30	498	43.6666
3	A-1f0ac7	HealthTech	UK	24	True	False	764.0	2500.682927	42.0	10	82	29.0000
4	A-ce550d	HealthTech	US	35	False	True	4053.0	3720.327586	217.0	28	406	42.2857

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7. Customer Health Score System

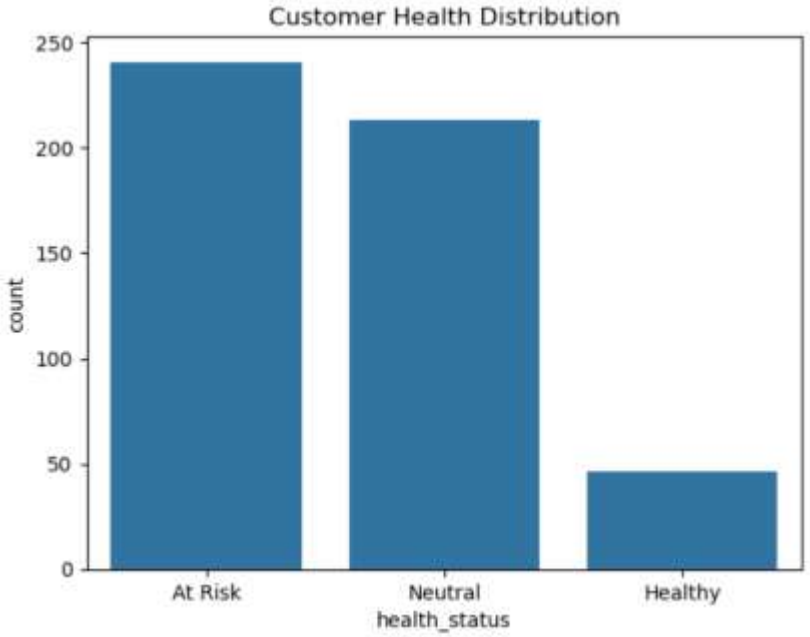
A health score was created using:

- Product usage
- Satisfaction

- Errors
- Support tickets

Segments:

- Healthy
- Neutral
- At Risk



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8. SQL Analysis

Key Queries Performed:

1. Churn Rate by Industry

- Identified industries with high churn

2. Support Impact on Churn

- Higher support tickets → lower satisfaction → higher churn

3. Customers Likely to Churn

- Based on usage, errors, and support metrics

	industry text	total_customers bigint	churned_customers bigint	churn_rate_percentage numeric
1	DevTools	113	35	30.97
2	FinTech	112	25	22.32
3	HealthTech	96	21	21.88
4	EdTech	79	13	16.46
5	Cybersecurity	100	16	16.00

4. High Value Customers

- Based on account_id, industry, total_arr

	account_id text	industry text	total_arr numeric
1	A-e08cd3	FinTech	62609088
2	A-0cc442	EdTech	57150432
3	A-1e1b80	FinTech	53382456
4	A-68f37c	HealthTech	50237040
5	A-180abf	DevTools	48971520
6	A-2d1036	FinTech	41656680
7	A-c58f49	EdTech	38616480
8	A-5b1bcd	DevTools	38466912
9	A-76fa4d	HealthTech	36986760
10	A-526d93	HealthTech	33219720

5. Feature Engagement Analysis

	industry text	avg_feature_usage numeric
1	HealthTech	3107.8750000000000000
2	EdTech	3031.0759493670886076
3	FinTech	3026.7410714285714286
4	Cybersecurity	2926.1900000000000000
5	DevTools	2733.6283185840707965

9. Key Data Insights

- Overall Churn Rate: ~22%
- Majority customers fall into "At Risk" category
- Low usage customers show higher churn probability
- High error rates strongly correlate with churn
- Customers with more support tickets show lower satisfaction

10. Key Learning

- Learned feature engineering (event → customer level)
- Understood importance of aggregation in analytics
- Learned how to combine SQL + Python + BI tools

11. Conclusion

This project transformed raw SaaS event-level data into a structured customer-level dataset using data cleaning, merging, and feature engineering. Key metrics like usage, errors, support activity, and revenue were derived to enable meaningful analysis.

The final dataset helped identify important patterns such as low engagement and high error rates leading to churn. A Customer Health Score was also created to segment users into Healthy, Neutral, and At Risk categories.

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Tools Used: Python | SQL | PowerBI | Jupyter Notebook

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